

An Area/Power-Efficient Noise-Shaping SAR ADC for Implantable Biosensor Applications Featuring a Unique Auxiliary Feedback Loop

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Abstract—Wide bandwidth and power-area efficient front-ends are required in emerging implantable biosensor applications, e.g., wireless artificial vision systems for the limited vision. This work presents a low-power 0.031 mm² 3-MHz bandwidth noise-shaping (NS) successive approximation register (SAR) analog-to-digital converter (ADC) tailored for implantable biosensor applications. A highly energy- and area-efficient reset-free residue-processing scheme is proposed. This scheme facilitates noise transfer function (NTF) optimization by enabling control over complex poles. It establishes a unique auxiliary feedback path from the infinite impulse response (IIR) back to the finite impulse response (FIR) by eliminating the need for the FIR filter's reset phase. This auxiliary feedback loop, which could not be achieved in previous FIR-IIR structures, is a key enabler for achieving high energy efficiency and design flexibility. The in-band quantization noise suppression is boosted by the proposed complex conjugate poles optimization technique, while a low kT/C input-referred noise has resulted. The prototype, fabricated in 65-nm CMOS technology with a 7-bit digital-to-analog converter (DAC), consumes 370 μ W from a 1-V supply and occupies an active area of only 0.17 $\times$ 0.18 mm (0.031-mm²). Operating at an 80-MHz oversampling frequency, the proposed ADC achieves a signal-to-noise and distortion ratio (SNDR) of 70.2 dB over a 3 MHz bandwidth, demonstrating its small area, high efficiency, and high performance for implantable biosensor applications that require high data rates.

Index Terms—Analog-to-digital converter (ADC), implantable biosensor applications, low area and power consumption, noise shaping (NS), stability, successive approximation register (SAR).

I. INTRODUCTION

THE first implantable medical biosensor devices were developed in the late 1950s with the advent of pacemakers [1], [2] and subsequent cochlear implants. At present, the development of personal telemedicine systems and the Internet of Medical Things (IoMT) drives the need for intelligent bio-electrical signal systems such as those used in biosensor applications with strict size and power constraints. Consequently,

the research community has shown great interest in developing low-power, area-efficient implantable biosensors and monitoring systems to assess physical health conditions. A typical readout front-end of an implantable biosensor system [Fig. 1(a)] involves sampling biological signals from the sensor, acquiring them by an analog front-end (AFE) or radio frequency (RF) transceiver, and converting them into digital codes to be processed by a digital signal processor (DSP). From the system perspective, the analog-to-digital converter (ADC) serves as the critical interface to acquire multi-channel biosignals in implanted biosensor devices (e.g., brain or peripheral nerve systems, etc. [3], [4]), enabling continuous monitoring and recording of biomedical signals with high resolution for accurate diagnosis and treatment of various medical conditions.

For implantable biosensor systems, an ADC featuring wide bandwidth to accurately capture high-frequency biomedical signals is imperative. Common biopotential signals recorded and analyzed in modern clinical practice, such as ExG signals [electroencephalogram (EEG), electrocardiogram (ECG), electromyogram (EMG), electrooculogram (EOG)], Axon Action Potential (AAP), retina, and optic nerve waves [5], [6], are shown in Fig. 1(a). In a low-pass filter (LPF) only transceiver without data downsampling, the ADC directly digitizes the analog signal without any prior signal processing, necessitating a conversion rate higher than the required data rate. In addition to wide bandwidth, it is essential to maintain low-power consumption to extend battery life in implantable devices and minimize the ADC area for improving comfort and biocompatibility [7]. In [8], a low-power 12-bit 1-kS/s SAR ADC for biomedical signal processing system, it achieves 10.47-ENOB with 110-nW power, while occupying an active area of 0.16 mm². In [9], a SAR ADC for bio-signal acquisition system is validated to enable the wireless measurement of EMG, ECG, and EEG with a power consumption of 1.22 μ W per channel, while operating within 0.2-2024 Hz. In addition, bandwidths in those designs [10], [11] are limited to less than 2-MHz, which is not adequate for high-data-rate biosensor applications such as endoscope capsules and retina implants. Similar to ultrasonic links demanding a data rate of 2 MS/s [12], the design of power- and area-efficient ADCs with high data rates has become critical for biosensor systems,

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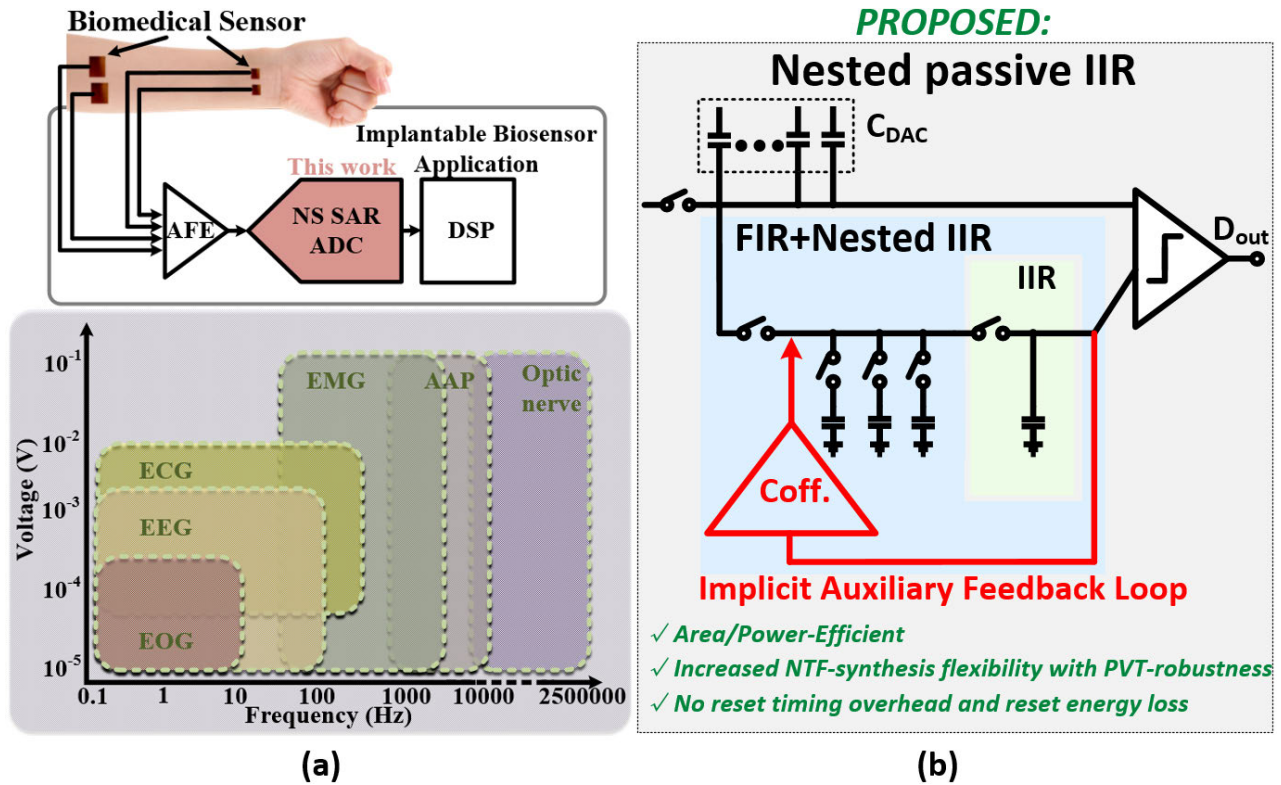


Fig. 1. (a) Typical implantable biosensor application and common biopotential signals. (b) Proposed nested passive IIR filter realizing reset-free NS with increased NTF-synthesis flexibility using an auxiliary feedback loop.

particularly for implantable biosensor applications, such as retina implants or neural recording requiring high data rates with small area [13].

Noise-shaping (NS) successive approximation register (SAR) ADCs have been developed, combining the NS technology of delta-sigma ($\Delta\Sigma$) ADCs with the low-power consumption advantage inherent in SAR ADCs [14]. NS SAR ADCs present a compelling architecture owing to their mostly passive nature, providing power and area efficiency for applications requiring moderate resolution and bandwidth [15], [16], [17], [18], [19], [20], [21], [22], [23], [24]. One effective strategy for enhancing noise transfer function (NTF) synthesis is to optimize the zero and pole locations [15], [16], [17]. Liu and Huang [15] introduced a finite impulse response (FIR) alongside a passive infinite impulse response (IIR) filter to enable NTF pole optimization; however, it requires a dynamic amplifier, which incurs additional power and is sensitive to process-voltage-temperature (PVT) variations. Oh et al. [22] employed one extra clock phase in each conversion to realize second-order passive IIR filtering for NTF zero optimization, but this method lacks the flexibility to adjust NTF poles, which can limit the achievable noise shaping performance. The NS SAR in [25] achieves a zero approximately at 1 on the z -plane with passive integration, but occupies a large active area because of the requirement of the quantization error storage capacitor. The technique of [19] achieves a sharp NTF, but requires a large integration capacitance, which often is not feasible for implantable biosensor ADC designs due to stringent area constraints.

This article presents a new NS SAR ADC to address several challenges in implementing NS SARs with limited power and area budgets for implantable biosensor applications. First, it incorporates a passive nested IIR filter [Fig. 1(b)] to facilitate complex-NTF-pole optimization and enhance the NTF-synthesis flexibility without necessitating an amplifier, leveraging an auxiliary feedback loop to achieve better NS performance. Second, it circumvents the reset phase (or any extra clock phase) commonly found in reported passive FIR-IIR NS SARs, which compromises power efficiency or operation speed. Third, it reduces the input-referred kT/C noise; therefore, smaller capacitors can be used under a given thermal noise budget. Fabricated in 65 nm CMOS and utilizing a 7-bit SAR, this prototype achieves a 70.2-dB signal-to-noise and distortion ratio (SNDR) across a 3 MHz bandwidth with an oversampling ratio (OSR) of 13.3, suitable for biosensor applications that require high conversion rates. The proposed NS SAR ADC yields a Walden figure of merit (FoMw) of 23.3 fJ/conv.-step, indicating high-energy efficiency, while occupying an area of only 0.031 mm², enabling compact integration for implantable biosensor systems. The contributions of this work include the formation, analysis, and silicon validation of the first NS SAR with a nested-IIR residue-processing architecture that features a reset-energy-loss-free scheme and achieves a second-order NS with a superior energy efficiency.

The article is organized as follows. Section II discusses the system architecture of the proposed reset-free NS SAR ADC. Section III presents the design considerations of the

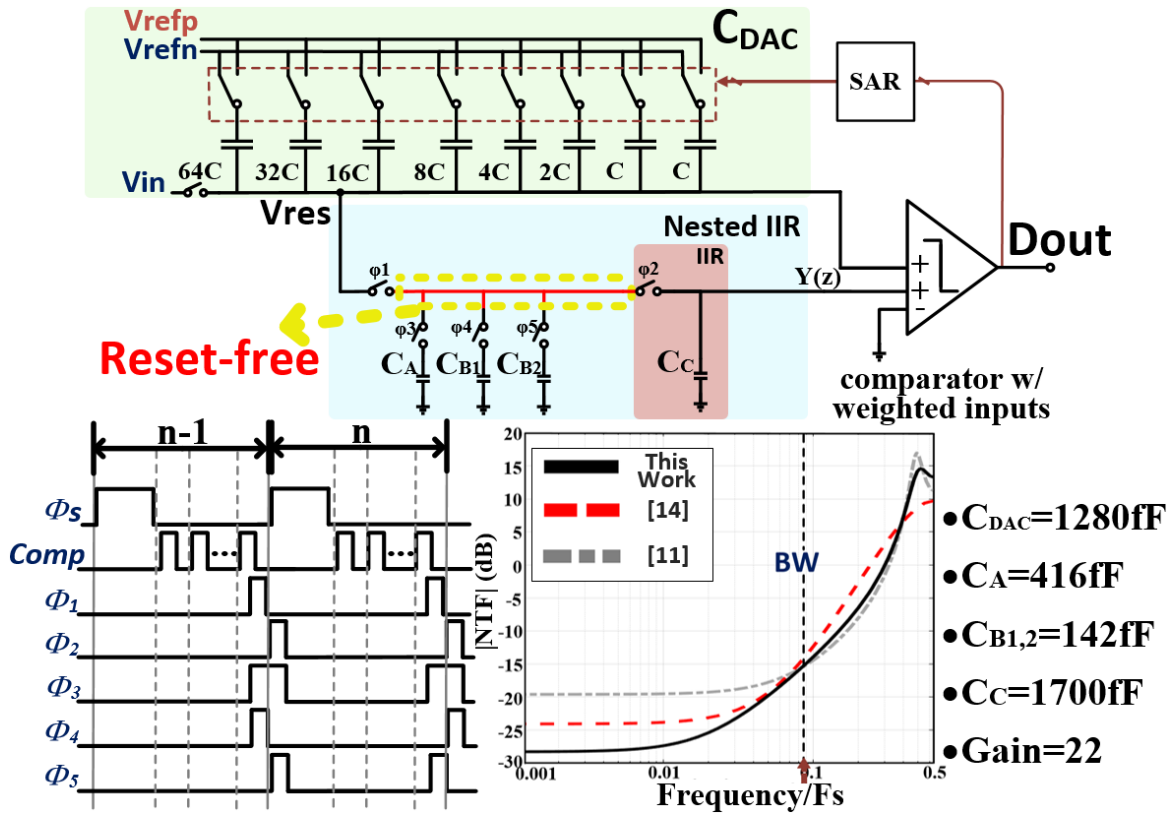


Fig. 2. Top level schematic and timing diagram of the proposed reset-free NS-SAR ADC architecture with the auxiliary feedback loop.

proposed nested passive IIR residue processing filter, stability and mismatch analysis, noise analysis of the proposed ADC, and comparator design trade-offs. The experimental results are given in Section IV. A conclusion is drawn in Section V.

II. PROPOSED NOISE-SHAPING ARCHITECTURE AND NESTED IIR RESIDUE PROCESSING

A. Proposed Noise Shaping Architecture

Fig. 2 illustrates the top-level circuit schematic and timing diagram of the proposed NS SAR ADC for implantable biosensor applications, where power efficiency and area constraints are critical considerations. A single-ended schematic is shown for simplicity, whereas the actual implementation is differential. C_{B1} and C_{B2} have the same nominal value of C_B . The SAR core uses a capacitive digital-to-analog converter (CDAC) configured to 7-bit resolution. The featured NS SAR incorporates a passive switched-capacitor (SC) filter that comprises an FIR + nested IIR section and an IIR section to process the residue voltage in the CDAC.

With this residue-processing scheme, the charges stored in the capacitors are not dumped, which inherently benefits the power efficiency. This residue-processing scheme, which will be further elaborated in Section II-B, initiates an auxiliary feedback loop within the circuit as depicted in Fig. 1(b). It thereby grants added flexibility in manipulating the resulting complex NTF poles, which is not available in the conventional implementations that achieve FIR-IIR NS. The auxiliary feedback loop is formed implicitly by the SC operation and improves the performance without hardware overhead. By this

property and the proposed optimization approach to be presented in Section III, the in-band noise shaping is enhanced with robust stability. The output of the residue processing circuit is forwarded to a comparator input, which has an embedded gain factor of A relative to its other input for V_{res} .

Conventional FIR-IIR implementations, e.g., [15], involve clearing their memories stored in the capacitors (C_A and C_{Bi} in our case) in the residue processing circuit every time before the next conversion phase arrives. The pivotal feature of the proposed residue processing filter is that charges in the capacitors C_A , C_{B1} , and C_{B2} will not be reset between phases. Their charge memories will be shared with the residue charge on the C_{DAC} in the next conversion cycle. This configuration implements a distinct auxiliary feedback path [Fig. 1(b)], which will be detailed in the next section. Reset-free operation means that no dynamic amplifier is needed before the passive filter. In this work, a comparator with two input pairs is adopted to fulfill the achievable sequence of block placement. In this way, the proposed residue processing circuit, which has a set of passive circuit elements identical to that in [15], realizes a better NS and thermal noise performance. An optimization method of the resulting complex NTF poles is also proposed. Under the chosen OSR, the proposed NS SAR improves the in-band noise suppression compared with the mainstream emerging arts [11], [14], as shown in the lower right corner of Fig. 2.

As depicted in the timing diagram in Fig. 2, the conversion process is divided into three sessions: input sampling, SAR conversion, and residue processing. The interval of the sam-

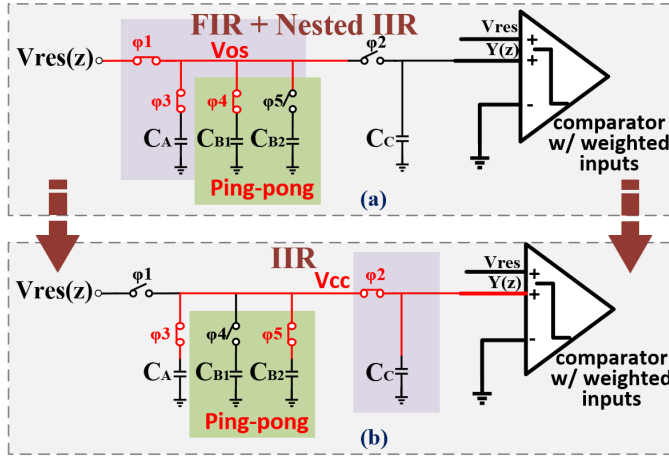


Fig. 3. Illustration of the reset-free residue-processing operations. (a) FIR + Nested IIR. (b) IIR.

pling session is spaced so as to meet the settling requirement to a 12-bit precision, which exceeds the target performance of the overall NS SAR. A bootstrapped switch is used to connect V_{in} to a common terminal of all the DAC array top plates for linearity and speed. The V_{cm} -switching scheme is adopted. During the SAR conversion, the residue processing circuit is disconnected from the C_{DAC} , ensuring that its capacitors do not share the signal charge on the C_{DAC} .

The inherent signal attenuation in the passive residue processing circuit due to the charge sharing among its capacitors is an important issue of any passive NS SAR architecture. This effect degrades the NS performance. Considering the aforementioned block placement sequence and power consumption, the proposed architecture employs a gain-embedded comparator to compensate for the signal attenuation, as shown in Fig. 2. The embedded gain is relative to the other input of the comparator and is determined by a transistor size ratio. The gain can thus be better controlled against PVT variations than the gain of a conventional dynamic comparator (e.g., [11]). To further optimize timing and power consumption and eliminate the need for an excessively fast external reference clock input, the SAR conversion phases are generated using an asynchronous clocking scheme [26]. In this scheme, a master clock at the oversampling rate controls the sampling instance, and a single delay element is used to time each digital-to-analog converter (DAC) settling event.

B. Nested IIR Residue Processing and NTF

In this section, we derive the NTF and construct a linear model of the proposed NS SAR ADC. For convenience in our analysis, we first define five capacitance ratios as follows: $a = C_A/(C_A + C_B + C_C)$, $b = C_{B1}/(C_A + C_B + C_C)$, $c = 1 - a - b$, $m = C_{DAC}/(C_{DAC} + C_A + C_B)$, $n = 1 - m$.

Fig. 3 details the operation of the passive residue-processing circuit, with the timing diagram for the switches provided in Fig. 2. Within the FIR and nested IIR section, C_A is used for a one-cycle delay (z^{-1}), whereas C_{B1} and C_{B2} serve the two-cycle delay (z^{-2}). The voltage on C_A , denoted as V_{OS} , is fed to the IIR section. The voltage on C_C , labeled as

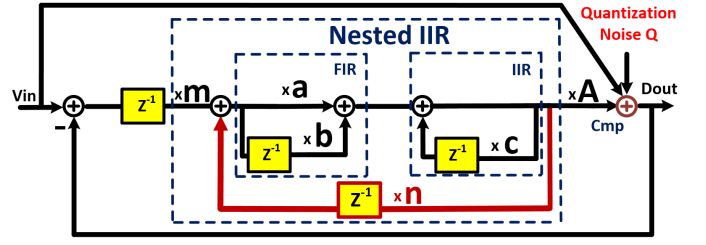


Fig. 4. Highlighted linear model of the proposed NS SAR ADC.

V_{CC} , is fed to the comparator. During the FIR and nested IIR operation phase, ϕ_1 goes high, and ϕ_3 and ϕ_4 (or ϕ_5 in alternate cycles) are also high. The residual charge on the CDAC (i.e., the residue from the prior SAR conversion cycle) is distributed between C_A and C_{B1} (or C_{B2}). The V_{OS} at the end of ϕ_1 can be expressed in the z -domain as

$$V_{OS} = m \cdot V_{res} + n \cdot V_{CC} \cdot z^{-1} \quad (1)$$

where V_{res} is the residue voltage on C_{DAC} after a conversion.

In the subsequent IIR operation phase, ϕ_2 goes high, and ϕ_3 and ϕ_5 (or ϕ_4 in alternate cycles) are also high. The charge on C_A and the charge on C_{B2} (or C_{B1}), which stores the V_{OS} of the previous conversion cycle, are both shared with C_C . The V_{CC} at the end of ϕ_2 can thus be expressed as

$$V_{CC} = \frac{a + b \cdot z^{-1}}{1 - c \cdot z^{-1}} \cdot [m \cdot V_{res} + n \cdot V_{CC} \cdot z^{-1}] \quad (2)$$

V_{CC} , after amplification by a factor of A , will then be used as the comparator's reference level (negated) for the SAR conversion in the following cycle. Hence, the transfer function of the residue filter in the proposed NS SAR is

$$H(z) = \frac{A \cdot z^{-1} \cdot V_{CC}}{V_{res}} = \frac{A \cdot m \cdot (a \cdot z^{-1} + b \cdot z^{-2})}{1 - (a \cdot n + c) \cdot z^{-1} - b \cdot n \cdot z^{-2}}. \quad (3)$$

Hence, the NTF of the proposed NS SAR is

NTF

$$\begin{aligned} &= \frac{1}{1 + H(z)} \\ &= \frac{1 - (a \cdot n + c) \cdot z^{-1} - b \cdot n \cdot z^{-2}}{1 + (A \cdot a \cdot m - a \cdot n - c) \cdot z^{-1} + (A \cdot b \cdot m - b \cdot n) \cdot z^{-2}}. \end{aligned} \quad (4)$$

All the coefficients in the NTF above are determined by capacitance ratios or a transistor ratio (for the gain A), which are immune to PVT variations.

Capturing the aforementioned passive nested IIR operations, Fig. 4 presents a linear model of the proposed NS SAR ADC. Owing to the charge memories reset-free configuration, an additional auxiliary feedback loop with coefficient n emerges, as highlighted in red in Fig. 4. In this case, the order of the loop filter (and thus the NTF) is increased by one.

From (4), the resulting complex poles of the proposed NS SAR ADC are $(x \pm yi)$, where the real and

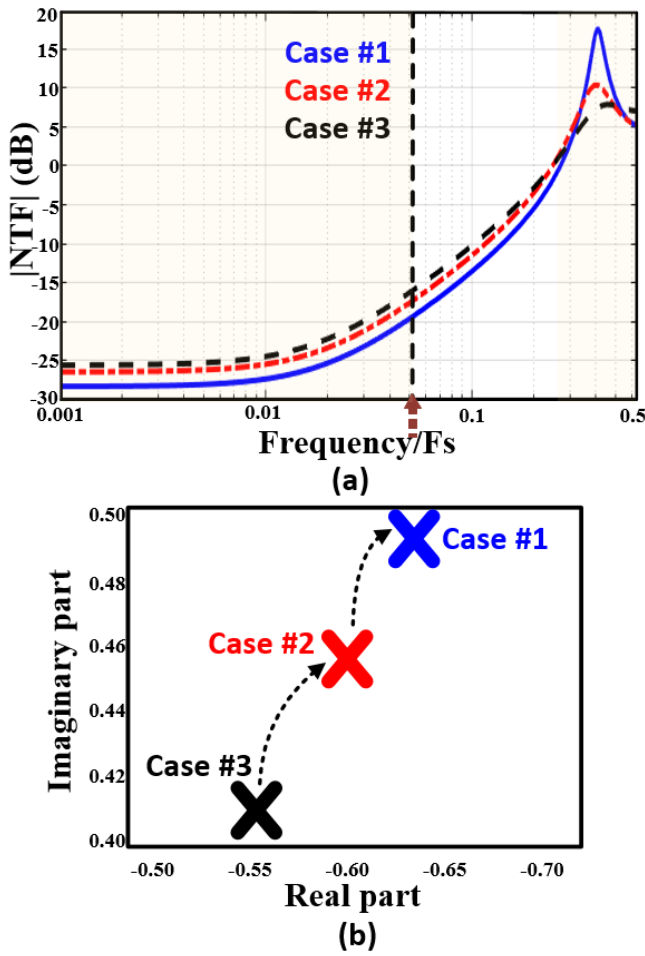


Fig. 5. (a) Magnitudes of the NTF. (b) Pole locations of the NTF.

imaginary parts are

$$x = -\frac{(A \cdot a \cdot m - a \cdot n - c)}{2}$$

$$y = \frac{\sqrt{4 \cdot (A \cdot b \cdot m - b \cdot n) - (A \cdot a \cdot m - a \cdot n - c)^2}}{2}. \quad (5)$$

These poles are a complex conjugate pair positioned within the unit circle on the complex plane, offering greater flexibility to adjust the poles. There is a sufficient degree of freedom to tune the pole location, which can be utilized for optimal suppression of the quantization noise within the bandwidth. This flexibility and adaptability are pivotal for the optimal suppression of quantization noise within the bandwidth. Fig. 5 shows the NTF magnitude responses of three pole cases. For example, in the case of Pole 1 ($-0.63, 0.49i$), it has a larger imaginary-part-to-real-part ratio compared with the cases of ($-0.60, 0.458i$) and ($-0.55, 0.41i$), and therefore the NTF has a flatter response and greater attenuation within the signal band. A more aggressive in-band attenuation of the NTF tends to increase its gain at high frequencies according to the Bode sensitivity integral. The NTF exhibits a more uniform response and enhanced suppression within the signal band when the poles possess a higher imaginary-part-to-real-part ratio considering stability.

C. Stability Analysis

Stability analysis of an NS SAR is essential for evaluating its performance. Left-half-circle poles can enhance the in-band noise attenuation of an NS SAR by enlarging the out-of-band (OOB) gain of the NTF. However, an NS SAR with left half-circle poles may compromise system stability due to the large peak in the NTF. To ensure the stability and robustness of the proposed NS SAR ADC while maintaining aggressive noise-shaping performance, a comprehensive stability analysis and optimization approach is necessary.

A restricted input range of the quantizer raises concerns about signal-dependent stability. Quantization error can generally be modeled as a uniformly distributed, additive, white noise source, which is accurate when the input signal does not overload the quantizer. The increased gain of the NTF at $\omega = \pi$ amplifies quantization noise to a greater degree as it propagates through the loop. An aggressive NTF, which provides deep suppression of in-band quantization noise, often has a limited stable input range.

According to the NTF of the proposed NS SAR ADC in (5), the zeros are determined by the capacitance ratio, whereas the poles are determined by both the capacitance ratio and the comparator gain A . Because the capacitance ratio is robust against PVT variations, we can substitute the capacitance coefficients into the NTF, leaving the comparator gain A as a variable. For the design presented here, the NTF is given by

$$\text{NTF}(z) = \frac{1 - 0.807z^{-1} - 0.0189z^{-2}}{1 + (0.133A - 0.807)z^{-1} + (0.0441A - 0.0189)z^{-2}}. \quad (6)$$

To maintain the stability of the ADC in this case, the poles of (6) are located within the unit circle. According to this condition, the stability condition decided by the comparator gain is as follows:

$$2.3454 \leq A \leq 31.6022. \quad (7)$$

Regarding the PVT variations, the adopted A is well within the stable region, which is far from the boundary condition. The stability of the proposed NS SAR ADC is further validated by the following mathematical analysis.

To evaluate the stable system conditions, the well-known method outlined in [27] has shown

$$|u|_{\max} = \max_n |u[n]| \leq N_{\text{lev}} + 1 - \sum_{k=0}^{\infty} |h[k]| \quad (8)$$

$$\sum_{k=0}^{\infty} |h[k]| \equiv \|h\|_1 \quad (9)$$

where $|u|_{\max}$ is the input MSA, N_{lev} is the number of quantization levels (i.e., the number of quantization thresholds plus 1), $h[k]$ corresponds to the impulse response, and $\|h\|_1$ is the L1-norm of the overall impulse response. The resolution of the NS SAR ADC is adopted 7-bit in this design ($N_{\text{lev}} = 128$). The impulse response $h[k]$ and consequently $\|h\|_1$ are determined by the placement of the complex conjugate NTF poles and zeros. From simulations in Simulink, the NS SAR achieves

an MSA of 0 dBFS. The circuit-level validation shows that the input range of the quantizer does not exceed the limit, and the linear model shown in Fig. 4 is accurate.

The stability can be cross-checked by examining $\|H\|_\infty$. The limit of $\|H\|_\infty$ is mathematically justified by comparing the norms via the z -transform of the impulse response $h(n)$ using the following formula:

$$\max \left(\left| \sum_{n=0}^{\infty} (h(n) \cdot e^{-j\omega n}) \right| \right) \leq \sum_{n=0}^{\infty} (|h(n)|) \quad (10)$$

$$\|H\|_\infty \leq \|h\|_1. \quad (11)$$

In this case, the achieved $\|H\|_\infty$ with the adopting case of Pole 1 shown in Fig. 5 does not exceed the required infinity norm. The stability criterion given in (10) and (11) is less pessimistic than that given in (8) and (9). Mathematically, the proposed architecture is feasible.

For a more balanced view of the proposed architecture, the potential limitations are discussed here. Although there is no restriction on the OOB spectral shape of NTF, the maximum magnitude of the NTF across all frequencies ($\|H\|_\infty$) should be considered to ensure stability, which involves a trade-off with noise shaping effect. While optimizing the introduced complex poles for noise shaping, the feasibility of the resulting NTF should be verified through behavior and PVT simulations. As for the mismatch issue, a mismatch between the two C_B 's in the residue processing circuit could cause nonlinearity. Simulations show that the degradation in the SFDR is less than 1 dB with a mismatch within $\pm 1.5\%$. Because the C_{B1} and C_{B2} are placed closely, their actual mismatch can be much less than 1.5%. Therefore, the performance degradation due to the mismatch is negligible.

D. Design Choice

We set constraints for the real and imaginary parts to find an optimal solution. The cases shown in Fig. 5 are examples. Oriented from the performance and stability concerns, initially, we set the radius of the complex pole to be within 0.9. Second, we calculate the $\|h\|_1$, the L1-norm of the overall impulse response. A smaller $\|h\|_1$ indicates a larger input MSA [shown in (8)]. The SNDR performance is the main concern during the design. The resulting SQNR should be larger than 80 dB. Setting optimization constraints for the complex poles is to limit the non-stable pole results. The trade-offs between the SNDR and input MSA exist during the optimization. We finally choose complex pole pairs $(-0.63, \pm 0.49i)$ as the design parameter, because the resulting NTF has flatter and better in-band noise suppression, and the MSA is relatively high. A fine-tuning of the complex poles is required during the post-layout because of the parasitic capacitance, and the expected NTF can be realized.

As one can see from (8), the signal-to-quantization noise ratio (SQNR) can be increased by using a larger path gain. A larger coefficient A makes the $\|H\|_\infty$ of the NS SAR larger, which makes the transfer function suffer from a higher $\|h\|_1$. The theoretical calculations and simulations show that the MSA degrades as A increases. A trade-off thus exists between the SQNR and the expected input range for different values

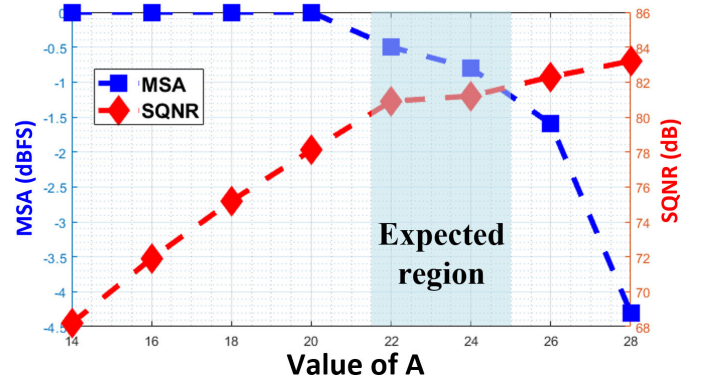


Fig. 6. MSA and SQNR versus A .

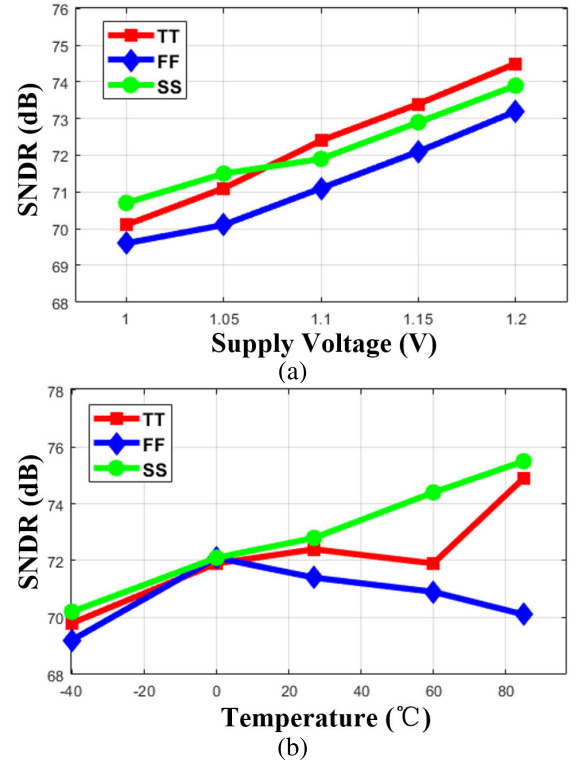
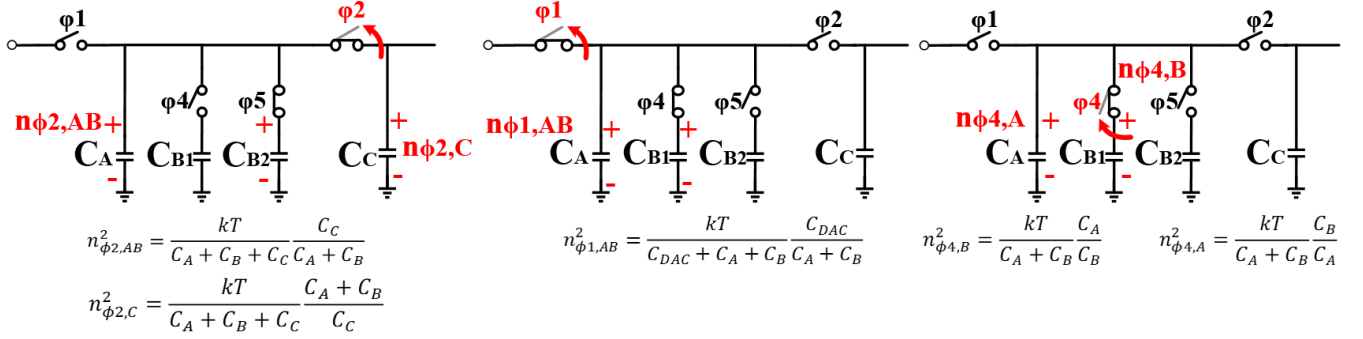


Fig. 7. Simulated SNDR versus temperature, voltage, and corner variations. (a) 1.0 to 1.2 V, FF, TT, SS. (b) -40°C to 85°C , FF, TT, SS.

of A , as shown in Fig. 6. The gain of the comparator should be chosen within the expected region indicated, in which the MSA is greater than -1.5 dBFS and the SQNR is maintained at a high level. Accordingly, the comparator gain is chosen as 22 in our design.

Fig. 7 illustrates the simulated SNDR performance of the NS SAR operating at 80-MS/s over temperature, voltage, and corner variations (-40°C to 85°C , FF, TT, SS, 1.0 V to 1.2 V). The SNDR performance in the design is post-simulated over a temperature range of $[-40^\circ\text{C}, 85^\circ\text{C}]$ for the biological safety concerns. Fig. 7(a) indicates that different supply voltages have little effect on the linearity of the ADC, with the SNDR exhibiting a variation of less than 4.4 dB. Fig. 7(b) demonstrates that the maximum change in SNDR is maintained at about 5 dB across the temperature range of -40°C

Fig. 8. kT/C noise in different operation phases.

to 85 °C for different corners. They provide additional insight into performance stability.

According to the post-layout extraction, the parasitic capacitance value of the cared nodes is obtained. A fine-tuning of the nested IIR capacitors (C_C , C_A , $C_{B1,2}$) is required during post-layout to fulfill the designed capacitance ratios of a , b , c , m , and n (defined in Section II-B), and the expected NTF can be realized. On the other hand, the IIR capacitance C_C is connected to the CDAC with two switches, resulting in a very small coupling capacitance. The parasitic capacitors between CDAC and C_C have little effect on the variation of the realized complex poles.

III. NOISE ANALYSIS AND BUILDING BLOCK CIRCUITS

A. Thermal Noise in the Proposed NS SAR ADC

Fig. 8 investigates the noise effects of the proposed NS-SAR ADC. The cumulative input-referred noise voltage is denoted as n_{o_total} , and $H_{ni}(z)$ represents the equivalent transfer functions from individual noise sources to the input.

The noise effects of the proposed NS-SAR ADC are investigated in Fig. 8. The total output noise is expressed as

$$n_{o_total} = \text{NTF}(z)[Q + n_{q1}] + H_{n1}(z)n_s + H_{n2}(z)n_{\phi 1, AB} + H_{n3}(z)n_{\phi 4, A} + H_{n4}(z)n_{\phi 4, B} + H_{n5}(z)n_{\phi 2, C} + H_{n6}(z)n_{\phi 2, AB} + H_{n7}(z)n_{q2} \quad (12)$$

where the transfer function $H_{n1}(z)$, $H_{n2}(z)$, $H_{n3}(z)$, $H_{n4}(z)$, $H_{n5}(z)$, $H_{n6}(z)$ and $H_{n7}(z)$ are

$$\begin{aligned} H_{n1} &= 1, & H_{n2} &= \frac{1}{(1-n)} \\ H_{n3} &= \frac{1}{(1-n)} \frac{a}{(1-c)}, & H_{n4} &= \frac{1}{(1-n)} \frac{b}{(1-c)} \\ H_{n5} &= \frac{1}{(1-n)} \frac{1}{(1-c)}, & H_{n6} &= \frac{n}{(1-n)} \frac{1}{(1-c)} \\ H_{n7} &= 1. \end{aligned} \quad (13)$$

n_{q2} represents the quantizer noise at the second input branch (A-branch). Obviously, the A-branch of the comparator noise n_{q2} is not suppressed in the signal band.

Q , n_{q1} represent the quantization error and circuit noise, respectively, at the first input branch. These two noises will be second-order shaped by the NTF of the system, therefore imposing an insignificant effect on the cumulative noise. n_s is the differential input sampling noise given by $(2kT/C_{DAC})$.

The terms $n_{\phi 1, AB}$, $n_{\phi 4, A}$, $n_{\phi 4, B}$, $n_{\phi 2, C}$, and $n_{\phi 2, AB}$ are sampled noise voltages in the SC operations phases, as indicated in Fig. 8. The derivation of their noise power during SC operation is shown in Fig. 8 accordingly. Treat C_A and C_B as one capacitor as they are shorted when ϕ_2 turns off. They still exchange energy after ϕ_2 turns off. The thermal noise sampled onto C_C and $C_A + C_B$ at the moment ϕ_2 turns off are as follows:

$$\overline{n_{\phi 2, C, in}^2} = \frac{kT}{(1-n)^2} \frac{C_A + C_B + C_C}{(C_A + C_B)C_C} \quad (14)$$

$$\overline{n_{\phi 2, AB, in}^2} = \frac{kTn^2}{(1-n)^2} \frac{C_C}{(C_A + C_B)^2}. \quad (15)$$

The thermal noise of C_{DAC} and thermal noise sampled onto $C_A + C_B$ at the moment ϕ_1 turns off are

$$\overline{n_s^2} = \frac{kT}{C_{DAC}}, \quad \overline{n_{\phi 1, AB, in}^2} = \frac{kT}{C_{DAC}} \frac{C_{DAC} + C_A + C_B}{C_A + C_B}. \quad (16)$$

ϕ_4 switches off before ϕ_2 and ϕ_5 turn on. The noise sampled onto C_A and C_B are

$$\overline{n_{\phi 4, B, in}^2} = \frac{kT}{C_A + C_B} \frac{C_A}{C_B} \frac{b^2}{(1-n)^2(1-c)^2} \quad (17)$$

$$\overline{n_{\phi 4, A, in}^2} = \frac{kT}{C_A + C_B} \frac{C_B}{C_A} \frac{a^2}{(1-n)^2(1-c)^2}. \quad (18)$$

This analysis serves as a detailed guideline for noise budgeting and choosing the NTF coefficients during the design of the proposed NS SAR ADC. A small n , defined at Section III-A, is preferred for a low-noise design. The results of this analysis offer insight into noise budgeting and the selection of NTF coefficients. Compared with the regular 7-bit SAR ADC, the proposed NS SAR can achieve a $1.75\times$ increase in the C_{DAC} capacitance under the same overall kT/C noise budget, making it an area-efficient choice for low-noise biosensor applications.

To facilitate noise analysis, a linear system diagram of the proposed NS SAR with thermal noises included is shown in Fig. 9.

B. Comparator

The comparator is a crucial building block in SAR ADC designs, significantly affecting the overall ADC performance in terms of accuracy, speed, and power consumption.

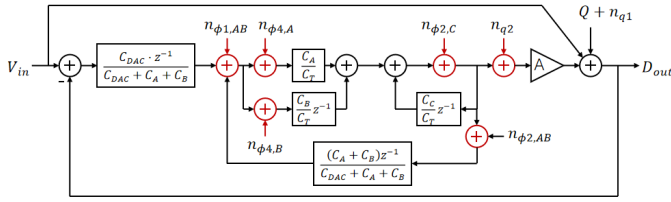


Fig. 9. Linear model of the NS SAR with thermal noises included.

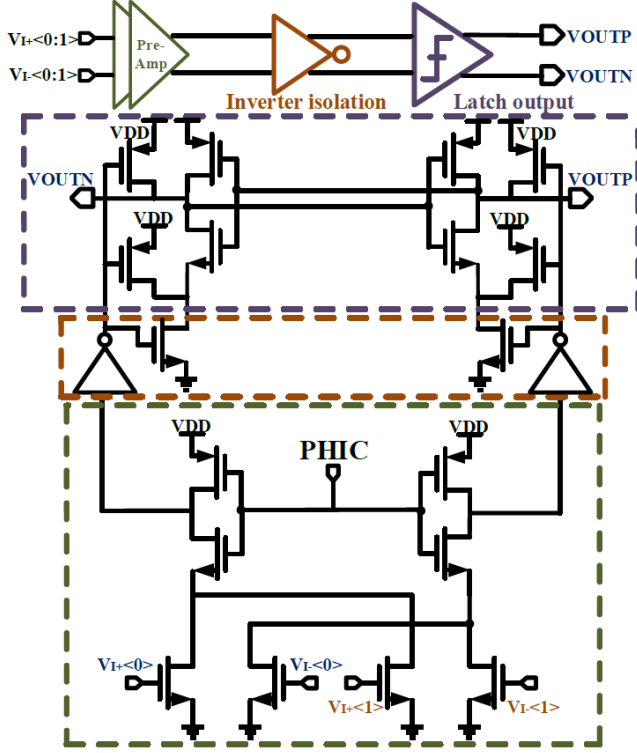


Fig. 10. Schematic of the low-noise three-stage comparator.

Dynamic latched comparators can achieve a high feedback gain more easily by incorporating a positive feedback load, which does not consume static power, making them widely used in low-power ADC designs. However, dynamic comparators can present issues of kickback noise and offset, which can degrade the ADC's SNDR and linearity, respectively. In the case of the StrongArm latch-based comparator used in [19], both the gate-source and gate-drain couplings to the input contribute to kickback noise, and the drains experience rail-to-rail transient signal transitions at clock-rising edges. To address the kickback noise and offset issues associated with dynamic comparators while retaining their low-power and high-gain advantages, the proposed NS SAR ADC incorporates a two-input low-noise three-stage comparator, as shown in Fig. 10.

In the comparator, the sources of the input pairs are grounded, thereby removing the kickback due to gate-source coupling. By grounding the sources, the comparator suffers only from the kickback resulting from the gate-drain coupling, and the drain voltage swings are only a fraction of the rail-to-rail voltage swing. As a result, the arrangement in the comparator with grounded input pair sources reduces the

kickback noise. In addition, it removes the dependence of the source voltage on the gate voltage, making the path gain A first-order independent of the input differential mode voltages. The width ratios of the two input differential pairs are set to 1:22, thereby realizing a relative path gain of $A = 22$, as illustrated.

Conventionally, a preamplifier stage is often used before the regenerative latch stage in comparator designs to reduce the input-referred offset. However, preamplifier-based comparators exhibit large static power consumption for achieving large bandwidth and reduced intrinsic gain due to the reduction in drain-to-source resistance (r_{ds}) with continued technology scaling [28]. Moreover, the offset of the preamplifier itself should also be considered. As the second stage, two inverters act as dynamic amplifiers and serve as an intermediate buffer between the first-stage pre-amplifier and the last-stage latch, reducing the loading on the first stage and accelerating the comparison speed. In addition, they provide extra voltage gain, reducing the noise and offset contributed by the latch stage, as well as shortening the time needed for latch regeneration. The outputs of the inverter stage are applied to the latch stage, which serves as both input and clock signals for the latch.

As shown in Fig. 2, the first input of the comparator inputs is connected to the V_{res} , which is connected to the plate of CDAC. The switching of the CDAC is a V_{cm} -based switching scheme, which will make the common-mode voltage of V_{res} (in the differential configuration) maintained at $V_{DD}/2$ as long as the common-mode voltage of V_{in} equals to $V_{DD}/2$. The common-mode voltage of the integration capacitor C_C is also $V_{DD}/2$ due to the difference structure. The common-mode voltage becomes $V_{DD}/2$ after around 100 runs. So, both inputs of the comparator have their common-mode voltages equal to $V_{DD}/2$.

These design choices, including grounding the sources of input pairs and incorporating an inverter-based intermediate buffer, not only speed the operation of the comparator but also reduce the kickback noise and static power, which can be favorable for low-noise power-constrained implantable biosensor applications.

Fig. 9 shows the linear system model of the NS SAR with all sources of noise included. If we consider the comparator noise at its output, it will be 22 times larger than that without using the additional input pair. There are three ways to refer the output noise of the comparator back to its inputs: First, we can refer all the comparator noise solely to the unity-gain input, i.e., with n_{q1} alone and $n_{q2} = 0$. Then, n_{q1} will be 22 times larger than the case without the gain path. However, n_{q1} will be shaped by the NTF the same way as the quantization noise Q is shaped. Thus, the in-band noise of the comparator when referred to the input of the NS SAR will be small. Second, we can refer all the comparator noise solely to the input of the gain path, i.e., with n_{q2} alone and $n_{q1} = 0$. Then, n_{q2} will be small as the noise power is attenuated by A^2 when referred back to the input of the gain path. Moreover, we can refer the noise generated from each input pair to the corresponding input of the comparator, i.e., the exact way as shown in Fig. 9. Then, n_{q1} is the same as that without using the additional input

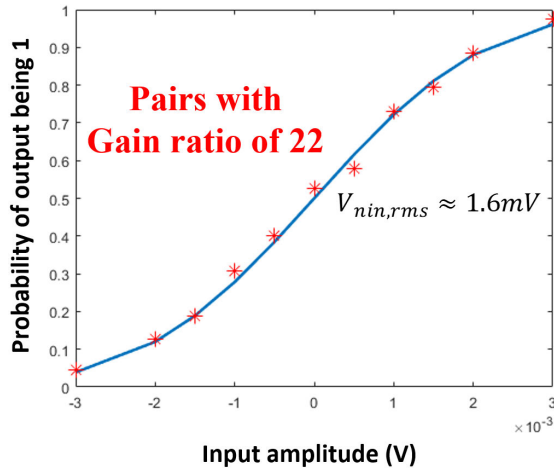
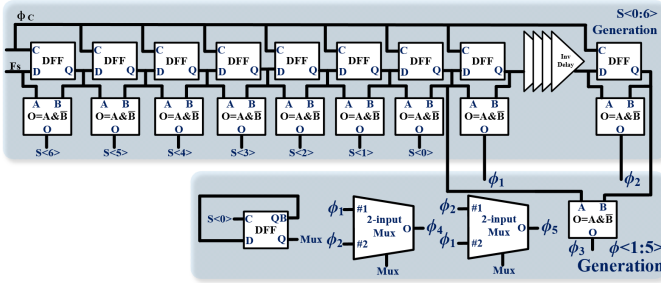


Fig. 11. Input referred noise of the input pair with the gain of 22.

Fig. 12. Pulses generation of $S<0:6>$ and $\phi<1:5>$.

pair and is shaped by the NTF. n_{q2} is small due to the large gain of the path. The input referred noise performance of the input pair with the gain of 22 are given in Fig. 11, following the cumulative distribution function (cdf). The above three approaches are equivalent and lead to the same conclusion that the addition of the gain path in the comparator has a negligible effect on the thermal noise performance of the NS SAR.

The offset of the comparator has little effect on the overall performance of the modulator because it is shaped by the NTF in the same way as the quantization noise is. According to the post-layout simulations, the offset of the comparator is 2-mV, which is negligible considering it is suppressed by the loop gain of the modulator in the signal band.

C. Logic Circuits

The conversion control pulses and passive integration control pulses are generated by the circuit, shown in Fig. 12. The inverter-based delay is used to make ϕ_1 and ϕ_2 non-overlapping. A DFF is used to generate a MUX control pulse with half of the sampling frequency. Two 2-input-MUX blocks are used here to generate ping-pong signals ϕ_4 and ϕ_5 . True-single-phase-clock (TSPC) is applied in the DFF design to reduce power consumption. The $A\&\bar{B}$ and MUX are dynamic logic circuits for the operation speed and the power efficiency.

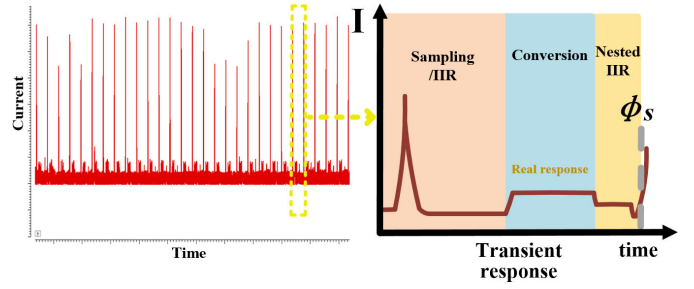


Fig. 13. Power consumption at different operation stages.

IV. MEASUREMENT RESULTS

The prototype of the proposed NS SAR ADC was fabricated in a 65 nm 1P9M general-purpose CMOS process. It occupies a compact active area of 0.031 mm². This compact active area is particularly noteworthy considering the excellent noise performance of the ADC achieved. From a 1-V supply, it consumes a mere 0.37 mW at an 80 MHz sampling frequency. As the transient response of the consuming current shown in Fig. 13, there are three phases when looking into the power consumption at different ADC operation stages, which are sampling/ IIR, conversion, and nested IIR stages. Initially, the bootstrapped switch is on to sample the V_{in} , which demands a spike current. The nested IIR operates within one clock cycle and consumes less power. The total consuming power is 0.37 mW under 1-V supply. The digital domain consumes 22% of total power. The comparator needs to output comparison results and generate the asynchronous clock and hence consumes a large portion of power. The nested IIR scheme utilizes the stored charge in the capacitors and contributes to the power efficiency of the residue processing operations. The die photo is shown in Fig. 14.

An FR4 printed circuit board (PCB) prototype was implemented to validate the proposed NS SAR performance targeted for implantable biosensor applications. As shown in Fig. 15, an input signal is provided by Rohde and Schwarz (R&S) SMA100B Signal Generator. At the external side of the test setup, the digital signal output coming from the testbench is captured and sampled by the employed R&S RTO2044 Digital Storage Oscilloscope for further processing on a PC. To emulate the intended implantable biosensor application during the pre-testing of the prototype, biopotential signals can be fed to the testbench via a frontend amplifier.

A -1.5 dBFS (where 0 dBFS = 2 V_{pp}, diff), 0.994 MHz sinusoidal signal from a source generator is fed to the external interface through a cascaded bandpass filter (Allen Avionics), which is utilized to attenuate the harmonics. The measured output power spectral densities (PSDs) of the NS SAR with NS enabled are shown in Fig. 16, with foreground DAC calibration. The PSD with the NS enabled clearly shows a second-order high-pass NS shape, which closely matches the theoretical NS shape shown in Fig. 5. The PSD is calculated from a Fast Fourier Transform (FFT) of the captured output data using a Blackman–Harris window to reduce spectral leakage. The measured peak SNDR and SFDR are 70.2 and 80.9 dB, respectively. The peak SNDR over the 3 MHz

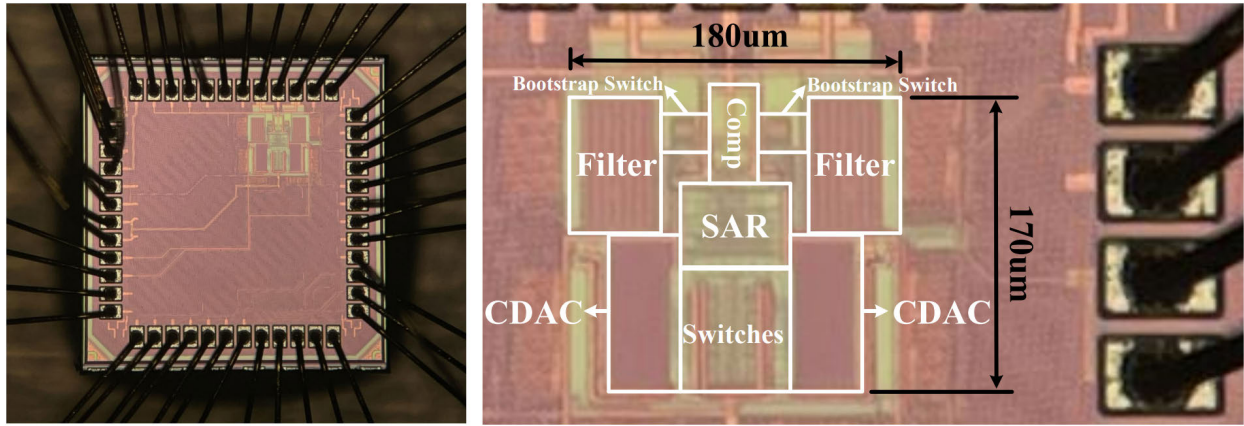


Fig. 14. Die microphotograph of the chip.

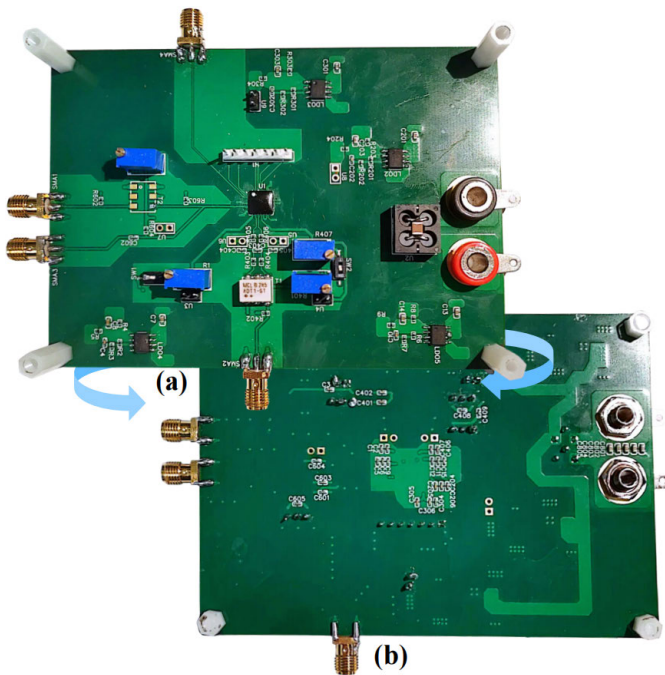


Fig. 15. PCB for testing (a) top view and (b) bottom view.

bandwidth improves to 70.2 dB with the NS on. The power distribution is detailed in the pie chart shown in Fig. 17. A foreground digital calibration, based on a least-mean-square (LMS) algorithm, is used to address the binary-weighted DAC mismatch issue. The calibration is based on an LMS algorithm, which generates the specified patterns with actual weights during foreground digital calibration. The same set of calibration coefficients is used during the measurement with different input frequencies and amplitudes. The corresponding Walden's FoM of the proposed design is 23.3-fJ/conv.-step, indicating its capability to perform efficient data conversions for the desired implantable biosensor applications. The spectrum with different input frequencies is shown in Fig. 18. The DSM function is validated with high-frequency inputs in Fig. 18(b).

Under an OSR of 13.3, this design achieves an 11.3-bit effective number of bits (ENOB) out of the 7-bit SAR core

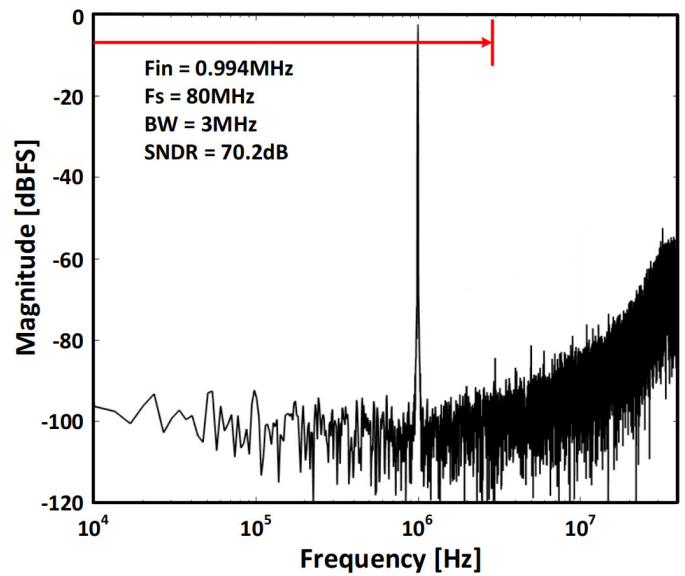
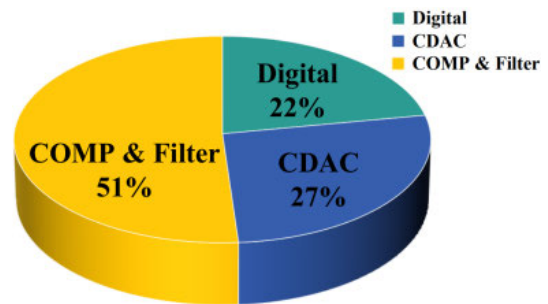
Fig. 16. Output spectrum of the ADC with a -1.5 dBFS input.

Fig. 17. Power breakdown.

thanks to the optimized NTF. The total required capacitance is only $2.75\times$ that of the 7-bit C_{DAC} . For comparison, a traditional 11-bit SAR ADC would require a capacitance of $16\times$ that of the 7-bit C_{DAC} , much larger than that in the proposed NS design. Because capacitors often dominate the chip area of a SAR ADC, the proposed NS SAR architecture can greatly reduce the chip area compared with the conventional ADC for implantable biosensor applications. This

TABLE I
PERFORMANCE COMPARISON

	[13]	[17]	[20]	[21]	[22]	This work
NS Method	Passive IIR	Active/Passive IIR	Passive FIR	NS Pipeline	Passive IIR	Nested Passive IIR
Area (mm ²)	0.081	0.17	0.072	0.38	0.113	0.031
Technology (nm)	65	65	65	180	65	65
Sampling Frequency (MHz)	80	10	10	24	83.3	80
OSR	20	20	8	8	7	13.3
Bandwidth	2	0.25	0.625	1.5	6	3
Peak SNDR (dB)	73.8	73	70.9	81.2	73.7	70.2
Power (mW)	2.13	0.094	0.13	1.8	3.5	0.37
FoM _s ^a [dB]	163.5	167.3	168	170.4	168.8	169.3
FoM _w ^b [fJ/c.-s.]	133	51.4	35.9	63.9	73.7	23.3

^bFoM_w = Power/(2×BW×2^Δ((SNDR-1.76)/6.02)).

^aFoM_s = SNDR + 10log₁₀(BW/Power).

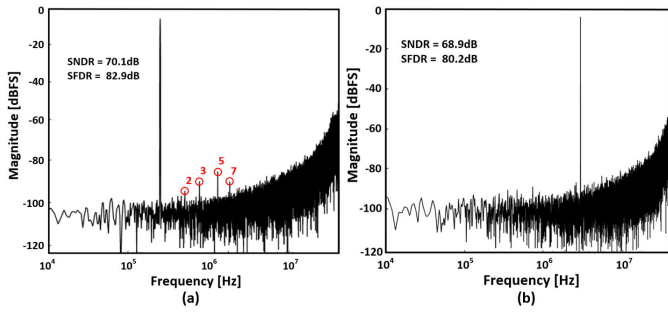


Fig. 18. Spectrum for different input frequencies. (a) Low-freq input. (b) High-freq input.

testifies that the hardware overhead required to implement our proposed pole-optimized NS SAR architecture is relatively low. Furthermore, the proposed technique is, in principle, capable of achieving higher order noise shaping, potentially enabling further performance improvements.

Table I summarizes the performance of the presented NS SAR ADC work compared to the state-of-the-art ADCs for the front-end considering comparable bandwidths and OSRs. Using the proposed nested IIR reset-free passive residue filtering technique, the ADC targeted for implantable biosensor applications achieves a high SNDR of 70.2 dB with a 3 MHz bandwidth. The power and area consumption of the ADC can be controlled to be 370 μ W and 0.031 mm². To the best of our knowledge, this chip realizes the smallest area considering comparable bandwidths and FoM_w with the same process. It shows that: 1) the reset-free nested-IIR-based NS filtering results in a high power-efficiency Walden's FoM of 23.3 fJ/conv.-step and 2) the thermal noise performance of the proposed architecture enables a small chip area.

V. CONCLUSION

This work is devoted to the development of a low-power area-efficient ADC with a low raw resolution for implantable biosensor applications. To achieve high-power efficiency and small area consumption for high-resolution data conversion

of biomedical signals, this article introduces a passive noise shaping technique featuring a reset-free nested-IIR architecture for residue processing with a crucial auxiliary feedback loop that enables high design flexibility. An 80 MS/s 3 MHz bandwidth 7-bit chip prototype was implemented and measured to validate the proposed method. The measured performance results conclude with an SNDR of 70.2 dB over 3 MHz and a power consumption of 370 μ W. The active chip area is miniaturized to 0.031 mm² (170 μ m × 180 μ m). The measured results validate the feasibility of the highly area- and energy-efficient scheme. Due to the reduced power consumption and core area achieved with the proposed techniques, the presented ADC architecture, along with the optimization techniques, is well-suited for implantable biosensor and wearable applications that require high data rates while operating under stringent space constraints, such as retina implants or neural recording [29] for biopotential signal detection.

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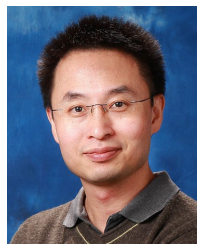
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